When 10-K Risk Text Adds to Accounting Ratios: A Text-Tabular Dataset for Next-Year Deterioration

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Abstract

Annual reports contain both accounting ratios and a long risk-factor discussion. We ask whether that text carries information about next-year financial deterioration beyond the ratios, and whether adapting the text encoder to the label adds anything further. We join SEC financial statements to Item 1A of the EDGAR corpus, draw 1,000 firm-years, and replace the risk excerpt with up to eight content phrases plus the short sentence that contains the first of those phrases. Five tabular learners are compared under structured, text-only, frozen-joint, and fine-tuned-joint conditions on five stratified folds. On the final table the mean joint gap is +0.020 AUROC and the mean fine-tuning gap is +0.012. All five learners clear both gaps. The joint gain is larger than the gain from ten lead sentences, and it is still split-sensitive.

1 Introduction

A Form 10-K states the same year twice: once as financial statements and once as prose. Accounting ratios have long been used to separate failing firms from healthy ones (Altman, 1968). The risk-factor section, Item 1A, is where management describes what could make the next year worse. Those two views are not the same object. A ratio records the current balance sheet. A risk sentence records a contingency that may not yet be visible in the accounts.

Text-tabular learning asks whether the prose still helps once the numbers are available. Arazi et al. (2026) separate that question into two measurable gaps. The joint gap is the gain of a frozen text-and-table model over the better single modality. The awareness gap is the further gain from steering the text encoder toward the prediction target. A useful dataset should show both, and should not do so only because the text restates a column that is already in the table.

This project builds that dataset for next-year deterioration and evaluates it with the official MulTaBench protocol. The contribution is the curation, not a new model. We keep ratios that describe size, leverage, liquidity, and turnover. The text column keeps up to eight phrases from the filing-specific Item 1A sentences, chosen on firm-years that are not in the reported draw, and appends the shortest sentence that contains the first phrase. Across five folds both average gaps are positive, and every learner clears both. The joint gap is about four times the gap of the earlier ten-lead table, and the fine-tuning gap is larger as well.

2 Related Work

Gradient-boosted trees remain strong tabular baselines. LightGBM grows trees leaf-wise (Ke et al., 2017). CatBoost reduces the bias of ordered boosting and handles heterogeneous columns (Prokhorenkova et al., 2018). TabM shows that a parameter-efficient ensemble of multilayer perceptrons can match more elaborate tabular networks (Gorishniy et al., 2025). TabPFN takes a different route: a transformer pretrained on synthetic tables performs in-context prediction on small datasets (Hollmann et al., 2025). TabPFN-2.5 extends that model to larger tables (Grinsztajn et al., 2025). MulTaBench evaluates exactly this mix, two tree models, TabM, and the two TabPFN versions, so that a dataset is not accepted because it suits only one learner (Arazi et al., 2026).

Unstructured inputs enter these models as embeddings. E5-small is a contrastively trained text encoder (Wang et al., 2022). A frozen encoder is optimized for general semantic similarity, so it can miss the wording that matters for one label. LoRA adapts a few low-rank weights without retraining the full network (Hu et al., 2022). MulTaBench applies that adaptation to the last three layers and calls the result a target-aware representation. The benchmark keeps a dataset only when, for at least three of five learners, the frozen joint model beats both unimodal models and the adapted encoder then beats the frozen joint model.

Financial text has its own line of work. Dictionary methods score the tone of a 10-K (Loughran and McDonald, 2011). Regression on the report text can predict market risk (Kogan et al., 2009). EDGAR-CORPUS supplies 10-K sections already split into items (Loukas et al., 2021). Those studies show that the filing contains predictive prose. They do not test whether that prose still helps after a small set of ratios is included, which is the comparison this dataset is built to support.

3 Method

3.1 Sources and join

Structured fields come from the SEC Financial Statement Data Sets, quarters 2015Q1 through 2021Q4, restricted to Form 10-K and 10-K/A with a full fiscal year. Text comes from EDGAR-CORPUS annual reports for 2016–2020 (Loukas et al., 2021). The tables are joined on the ten-digit CIK and the fiscal year. The 2021 statements supply the year-t+1 outcome for a 2020 filing, and 2015 supplies the prior year used to form changes. After the label filter below, 4,803 firm-years remain. We draw 1,000 of them uniformly with sample seed 42. The

draw is between firms and years, not a firm-held-out split.

3.2 Label

For each firm-year we form three changes from year t to $t+1$: operating income, interest coverage, and operating cash flow. Each change is standardized on the pool, and the deterioration score is their mean. A lower score is worse. The worst 25% of the score is the positive class and the best 55% is the negative class. The middle 20% is discarded, so the label compares clear deterioration with clear health rather than neighboring percentiles. On the 1,000-row draw the positive rate is 0.305. The label is a relative tail of this pool, not a bankruptcy filing.

3.3 Risk-factor text

Item 1A is used instead of the opening of Item 7. The opening of the management discussion is dominated by safe-harbor language, whereas Item 1A is the section that lists risks. A full Item 1A is far longer than the 512-token encoder window, and its first paragraphs are often a legend shared across filings. We therefore take the first sentence of each paragraph and apply four filters. A sentence must have 8 to 42 words and a digit fraction below 0.12. Sentences that contain a fixed list of boilerplate phrases are removed, including forward-looking statements, the securities-act legend, and generic cyber, personnel, and tax language. Sentences copied into more than 15% of filings are removed, as are sentences whose content words appear, on average, in more than 15% of filings. Of what remains, the first ten sentences in document order are the fallback excerpt. The reported table replaces that excerpt when the filing contains phrases that a text model found predictive on the other firm-years. A TF–IDF logistic regression, with unigrams and bigrams, is fit only on the 3,803 firm-years left out of the seed-42 draw. Terms are ranked by absolute coefficient. A term is kept only if it contains a token of at least four characters outside a short list of function and boilerplate words, and only the 250 strongest such terms are eligible. Each evaluation filing is then scanned for those terms in its frequency-filtered Item 1A sentences. Up to eight non-overlapping matches are stored, longest first. If a filing matches none, it keeps the original ten leads. That fallback covers 189 of the 1,000 rows. The phrase list itself is four words at the median. The reported excerpt then appends the shortest frequency-filtered sentence that contains the first selected phrase, cut at 20 words. 811 filings receive that sentence. The median combined excerpt is 21 words. A phrase list alone left the frozen joint model strong and the fine-tuned model weaker for the tree learners, so the sentence is what the final table adds. The CSV stores the excerpt without an extra encoder prefix. MulTaBench adds the E5 passage prefix itself.

3.4 Structured columns

The numeric columns are log total assets, debt to assets, the current ratio, cash to assets, and asset turnover. Margins, coverage, return on assets, growth, and capital expenditure were not used as inputs. Those quantities either restate the changes inside the label or tend to be narrated in the same words as the risk text. Capital expenditure to assets and fiscal year were removed after they failed to improve the average of the two gaps across folds. The current ratio and asset turnover are heavy-tailed, so each is replaced by $\operatorname{sign}(x)\log(1+|x|)$, rounded to three decimals. Coarsening leverage and liquidity into tertiles, and drawing one year per firm, did not raise the five-fold text score, so the reported table keeps these signed-log ratios and the original 1,000 rows. The other ratios stay on their winsorized scale, with the sample clipped at the 1st and 99th percentiles before that transform. Deleting rows, by contrast, was not helpful. Removing high-leverage firms or the smallest firms made the ratios easier and the text worse. The final table therefore changes columns, not the set of firm-years.

3.5 Evaluation

We use the MulTaBench evaluator without hyperparameter search. The five learners are LightGBM, CatBoost, TabM,

Table 1: Five-fold mean AUROC on the final table. The excerpt is the phrase list plus one host sentence. Gaps use means rounded to three decimals.

Model	Struct.	Text	Frozen	TAR	Δ_J	Δ_A
LightGBM	0.599	0.570	0.620	0.628	0.021	0.008
CatBoost	0.601	0.589	0.628	0.642	0.027	0.014
TabM	0.523	0.516	0.543	0.568	0.020	0.025
TabPFN 2.0	0.570	0.559	0.583	0.592	0.013	0.009
TabPFN 2.5	0.569	0.562	0.590	0.594	0.021	0.004
Average	0.572	0.559	0.593	0.605	0.020	0.012

TabPFN 2.0, and TabPFN-2.5. Each is trained in four conditions: ratios only, text only, both with a frozen E5-small encoder, and both after LoRA adaptation of the last three encoder layers. LoRA uses rank 16, learning rate 10^{-4}, at most 50 epochs, patience 3, and weight decay 10^{-3}. Embeddings are mean-pooled and reduced by PCA to 30 dimensions, fit on the training split only. Each fold holds out 10% of the rows with stratification. The fold index k uses random state $42+k$, so fold 0 is the split used during development. Condition means are rounded to three decimals before subtraction. For a learner, the joint gap is the frozen joint score minus the better unimodal score, and the awareness gap is the fine-tuned joint score minus the frozen joint score. The reported averages are the means of those five learner gaps.

4 Results

Table 1 is the five-fold mean on the final table. The average joint gap is +0.020 and the average awareness gap is +0.012. Every learner has both gaps positive. LightGBM's frozen joint score of 0.620 exceeds its ratios by 0.021, and fine-tuning adds 0.008. CatBoost adds 0.027 and then 0.014. TabM adds 0.020 and then 0.025. TabPFN 2.0 and TabPFN-2.5 add 0.013 and 0.021 from the frozen text, and fine-tuning adds 0.009 and 0.004.

The same ratios with the original ten leads produced average gaps of +0.005 and +0.005, and three learners cleared both (Appendix A). Phrases alone, without the host sentence, raised the average joint gap to +0.019 but left the average awareness gap at +0.002, and only TabPFN 2.0 cleared both (Appendix A). The structured scores are unchanged across these tables, because the ratios were not.

The gaps are still not stable across folds. For LightGBM on the final table, text-only ranges from 0.501 to 0.614 and the structured score stays between 0.558 and 0.652 (Appendix A). Fold 0 remains the development split. An earlier table, before fiscal year was removed and before the signed log, had average gaps of +0.036 and +0.018 on that split and average gaps of −0.001 and +0.004 on all five folds.

The phrase list is what makes the frozen embedding specific, and the host sentence is what gives LoRA a residual to adapt. With the phrases alone, fine-tuning lowered LightGBM, CatBoost, and TabM. Putting the phrases and the original ten leads into one string removed most of the LightGBM joint gap. Keeping the phrases and adding only the one short sentence leaves every joint gap positive and turns every awareness gap positive. A second text column for that sentence, scored without fine-tuning, made LightGBM's joint gap −0.019, because its text-only score rose above the joint score. A finance-term lexicon, a shock label, and a one-year-per-firm redraw did not produce a more stable text score.

TabM still has the weakest structured score, 0.523. Its joint gap of 0.020 is a lift over that ratio model, and fine-tuning adds a further 0.025, the largest awareness gap of the five.

5 Conclusion

Item 1A phrases selected outside the evaluation draw, followed by the one sentence that contains the first phrase, add more information about next-year deterioration than the ten lead sentences they replace. The five-fold mean joint gap is +0.020 and the mean fine-tuning gap is +0.012. All five learners have both gaps positive, which clears the three-learner bar used by the benchmark. A result computed on the development fold is not a safe summary of this dataset.

The label is a percentile of three accounting changes, not a default or a delisting, and the 1,000 rows are a random draw rather than a split that holds out firms. The phrase

list was chosen using the complement of that draw, and the decision to keep it used the five reported folds, so the five-fold mean is not a completely untouched evaluation. Test folds contain about 100 rows, which is enough for the sign of a gap to flip. A next study should freeze the table before the reported folds and split by firm rather than by row.

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A Additional result tables

Table 2: Five-fold mean AUROC before removing fiscal year and before the signed log. Capital expenditure is already excluded. This is the table whose fold 0 looked strongest.

Model	Struct.	Text	Frozen	TAR	Δ_J	Δ_A
LightGBM	0.611	0.516	0.559	0.590	−0.052	0.031
CatBoost	0.628	0.541	0.611	0.594	−0.017	−0.017
TabM	0.542	0.548	0.587	0.572	0.039	−0.015
TabPFN 2.0	0.579	0.568	0.594	0.604	0.015	0.010
TabPFN 2.5	0.582	0.584	0.595	0.605	0.011	0.010
Average	0.588	0.551	0.589	0.593	−0.001	0.004

Table 3: Five-fold mean AUROC with the original ten Item 1A leads, the same ratios, and the same 1,000 rows. This is the table whose average gaps were +0.005 and +0.005.

Model	Struct.	Text	Frozen	TAR	Δ_J	Δ_A
LightGBM	0.599	0.516	0.580	0.581	−0.019	0.001
CatBoost	0.601	0.541	0.608	0.596	0.007	−0.012
TabM	0.523	0.548	0.555	0.569	0.007	0.014
TabPFN 2.0	0.570	0.568	0.595	0.605	0.025	0.010
TabPFN 2.5	0.569	0.584	0.590	0.603	0.006	0.013
Average	0.572	0.551	0.586	0.591	0.005	0.005

Table 4: Five-fold mean AUROC with the phrase list and no host sentence. The average gaps are +0.019 and +0.002, and only TabPFN 2.0 clears both.

Model	Struct.	Text	Frozen	TAR	Δ_J	Δ_A
LightGBM	0.599	0.597	0.625	0.619	0.026	−0.006
CatBoost	0.601	0.608	0.654	0.641	0.046	−0.013
TabM	0.523	0.558	0.579	0.569	0.021	−0.010
TabPFN 2.0	0.570	0.569	0.577	0.599	0.007	0.022
TabPFN 2.5	0.569	0.586	0.579	0.598	−0.007	0.019
Average	0.572	0.584	0.603	0.605	0.019	0.002

Table 5: Per-fold AUROC for LightGBM on the final table, illustrating the split sensitivity in Section 5. Scores are rounded to three decimals before the gaps.

Fold	Struct.	Text	Frozen	TAR	Δ_J	Δ_A
0	0.586	0.567	0.652	0.692	0.066	0.040
1	0.608	0.614	0.656	0.683	0.042	0.027
2	0.558	0.565	0.608	0.620	0.043	0.012
3	0.652	0.606	0.633	0.618	−0.019	−0.015
4	0.590	0.501	0.548	0.527	−0.042	−0.021